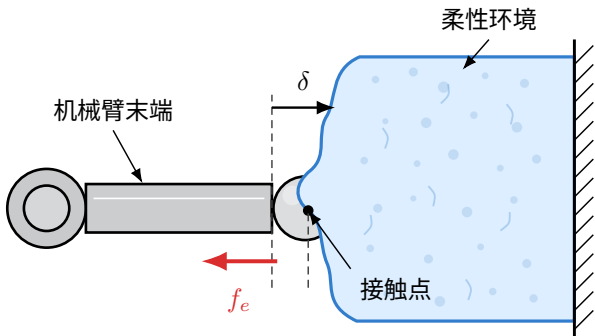
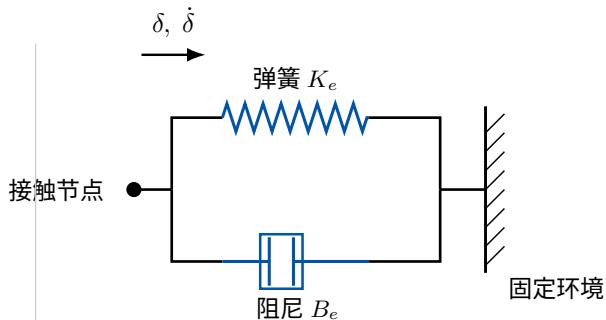




(a) 物理接触示意



$$f_e = K_e \delta + B_e \dot{\delta} + \Delta_f$$

(b) 等效力学模型

柔性环境接触建模的开尔文-沃伊特模型